Fiscal Year:	FY 2018	**Task Last Updated:**	FY 10/25/2017
PI Name:	Poirier, David Sc.D.		
Project Title:	Effect of Varying Convection on Dendrite Morphology and Macrosegregation		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	poirierd@email.arizona.edu	**Fax:**	FY 520-621-8059
PI Organization Type:	UNIVERSITY	**Phone:**	520-621-6072
Organization Name:	The University of Arizona		
PI Address 1:	1235 East James E Rogers Way		
PI Address 2:	Department of Materials Science and Engineering		
PI Web Page:	http://www.mse.arizona.edu/		
City:	Tucson	**State:**	AZ
Zip Code:	85721-0012	**Congressional District:**	7
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
Start Date:	12/01/2014	**End Date:**	11/30/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Grugel, Richard	**Contact Phone:**	256 544-9165
Contact Email:	Richard.N.Grugel@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	October 2017 report: Prof. D.R. Poirier, Dept. of Materials Science and Engineering, The University of Arizona, remains as the P.I. Dr. V. Manga, Assistant Research Professor, in the Dept. of Materials Science and Engineering at The University of Arizona has been the co-investigator since early 2016.		
COI Name (Institution):	Manga, Venkat Ph.D. (The University of Arizona) Tewari, Surendra Ph.D. (Cleveland State University)		
Grant/Contract No.:	NNX15AC04G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE (Ed., 12/16/2014): This project is a continuation of grant NNX10AV40G, "Effect of Varying Convection on Dendrite Morphology and Macrosegregation," with period of performance of 12/1/2010-11/30/2014. The research of this grant pertains to convection that is driven by directional solidification (DS) through changes in cross-sections (because of solidification shrinkage) and thermosolutal convection. In microgravity, of course, the thermosolutal convection will be mitigated so that effects of the shrinkage-driven flow in cross-sectional changes can be isolated from the usual thermosolutal convection that always occurs when solidification occurs on Earth. The outcome of the research is expected to be relevant to commercial DS-castings, which are used in the high-temperature sections of gas-turbine engines (i.e., jet engines and electric power generators). Such sophisticated castings have geometries, which invariably have changes in cross-sections. Yet convection at a cross-sectional change and its relationship to the formation of defects in DS-castings and/or in dendritic mono-crystals have not been studied well. Hence, the research not only has scientific merit, it is also relevant to the production of commercial DS castings. The research is on directional solidification experiments that are compatible with the solidification cartridge assemblies (SCAs) that can be employed in a furnace on the International Space Station (ISS). The research comprises: 1) selected alloys have varying density-responses to gravity forces; 2) crucible designs with variable cross-sections but compatible with the current SCAs; 3) sophisticated characterizations of the solidified microstructures, including both micro- and macro-segregation; and 4) supporting computer simulations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The outcome of the research is expected to be relevant to commercial directionally solidified-castings, which are used in the high-temperature sections of gas-turbine engines (i.e., jet engines and turbine electric power-generators). Such sophisticated castings have geometries, which invariably have changes in cross-sections. Yet convection at a cross-sectional change and its relationship to the formation of defects in DS-castings and/or in dendritic mono-crystals have not been studied well. Hence, the research is very much needed for its scientific merit and also because it is relevant to the production of commercial DS castings.
Task Progress:	Solidification simulations of two alloys have been completed; these include Al-7%Si and Al-19%Cu alloys. Both of these alloys have solutes that increase the density of the liquid as the alloys solidify, with the effect stronger in the Al-19Cu alloy. Experimental castings and the simulations show segregation patterns that agree. Computer simulations of the alloys solidifying in space predict no macrosegregation. When solidification takes place through an abrupt expansion, we also see evidence of so-called "stray grains" can form. For the remainder of the grant period effort will be on the directional solidification of Pb-6%Sb alloy. Unlike the two Al-alloys, this alloy has the characteristic that the density of the liquid decreases during solidification, and the macrosegregation pattern is expected to very different than that observed in the Al-alloys. For the computer simulations several/many physical properties are needed. These include diffusion coefficients in the liquid and the energy of the interface between the solid and liquid as the alloy solidifies. Some of our effort will be devoted to a critical examination of the available data of these properties and a reassessment of the same if needed.
Bibliography Type:	Description: (Last Updated: 06/06/2025)
Articles in Peer-reviewed Journals	Ghods M, Johnson L, Lauer M, Grugel RN, Tewari SN, Poirier DR. "Radial macrosegregation and dendrite clustering in directionally solidified Al–7Si and Al–19Cu alloys." Journal of Crystal Growth. 2016 May;441:107-16. http://dx.doi.org/10.1016/j.jcrysgro.2016.02.014 , May-2016
Articles in Peer-reviewed Journals	Lauer M, Ghods M, Angart SG, Grugel RN, Tewari SN, Poirier DR. "Macrosegregation during re-melting and holding of directionally solidified Al-7 wt.% Si alloy in microgravity." JOM (Journal of The Minerals, Metals & Materials Society (TMS). 2017 Aug; 69(8):1289-97. https://doi.org/10.1007/s11837-017-2380-0 , Aug-2017